Laura H. Yang

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EDUCATION

Harvard University — Cambridge, MA
Ph.D., Environmental Science and Engineering — 09/2021 - 08/2025

Georgia Institute of Technology — Atlanta, GA
B.S., Environmental Engineering — 09/2016 - 12/2020

PROFESSIONAL EXPERIENCE

Postdoctoral Research Fellow, Washington University in St. Louis — 09/2025 - Present
Host: Randall V. Martin

NSF Graduate Research Fellow, Harvard University — 09/2021 - 08/2025
Advisor: Daniel J. Jacob

NSF Intern Program Research Intern, NASA Jet Propulsion Laboratory — 02/2025 - 07/2025
Hosts: Kevin W. Bowman, Kazuyuki Miyazaki

Undergraduate Research Assistant, Georgia Institute of Technology — 05/2019 - 08/2021
Advisor: Nga L. (Sally) Ng

Global Internship Program Research Assistant, Yonsei University — 08/2019 - 08/2020
Host: Jhoon Kim

RESEARCH INTERESTS

I am an atmospheric chemist. I use chemical transport models, satellite data, and aircraft observations to study atmospheric composition, inform air quality policy, and protect human health.

HONORS, AWARDS, AND FELLOWSHIPS

Air Quality and Health Initiative Postdoctoral Fellowship, Washington University in St. Louis	2025
Intern Supplemental Funding, National Science Foundation	2025
Graduate Research Fellowship, National Science Foundation	2021 - 2025
Stonington Endowment Graduate Fellowship, Harvard University	2021
Sigma Xi Best Undergraduate Research Award, Georgia Tech	2021
Civil and Environmental Engineering Best Undergraduate Research Award, Georgia Tech	2021
Buck Stith Outstanding Senior Award, Georgia Tech	2021
President's Undergraduate Research Awards, Georgia Tech	2017, 2019
Buck Stith Outstanding Junior Award, Georgia Tech	2019
Callaway Scholarship, Georgia Tech	2017
Provost Scholarship, Georgia Tech	2016 - 2020
Outstanding Young Environmental Scientist Award, Cary Institute of Ecosystem Studies	2016

PEER-REVIEWED PUBLICATIONS

H-index: 13, citations: 455 (as of 06/2026, Google Scholar)

First-Author Papers

[19] **Yang, L.H.**, Jacob, D.J., Bates, K.H., Lin, H., H.M. Allen, J.F. Müller, Brown, S.S., Dang, R., Colombi N.K., Zhai, S., Yantosca, R.M., Brewer, J.F., Ng, N.L., Crounse, J.D., Wennberg, P.O., Li, K., Liao, H. Modeling of Methyl Hydroperoxide Observations in Urban and Remote Air over South Korea: Methyl Radical Chemistry and Inference of Atmospheric Methanediol. *Geophysical Research Letters.* 52(22), e2025GL118267, 2025. https://doi.org/10.1029/2025GL118267

[13] **Yang, L.H.**, Jacob, D.J., Lin, H., Dang, R., Bates, K.H., East, J.D., Travis, K.R., Pendergrass, D.C., Murray, L.T. Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. *Geophysical Research Letters*, 52(5), e2024GL112445, 2025.
https://doi.org/10.1029/2024GL112445

[9] **Yang, L.H.**, Jacob, D.J., Dang, R., Oak, Y.J., Lin, H., Kim, J., Zhai, S., Colombi, N.K., Pendergrass, D.C., Beaudry, E., Shah, V., Feng, X., Yantosca, R.M., Chong, H., Park, J., Lee, H., Lee, W.-J., Kim, S., Kim, E., Travis, K.R., Crawford, J.H., Liao, H. Interpreting Geostationary Environment Monitoring Spectrometer (GEMS) Observations of the Diurnal Variation in NO_2 over East Asia. *Atmospheric Chemistry and Physics*, 24(12), 7027–7039, 2024.
https://doi.org/10.5194/acp-24-7027-2024

[4] **Yang, L.H.**, Jacob, D.J., Colombi, N.K., Bates, K., Shah, V., Beaudry, E., Yantosca, R.M., Lin, H., Brewer, J., Chong, H., Travis, K., Crawford, J., Lamsal, L., Koo, J.-H., Kim, J. Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry. *Atmospheric Chemistry and Physics*, 23(4), 2465–2481, 2023.
https://doi.org/10.5194/acp-23-2465-2023

[3] **Yang, L.H.**, Hagan, D.H., Rivera-Rios, J.C., Kelp, M.M., Cross, E.S., Peng, Y., Kaiser, J., Williams, L.R., Croteau, P.L., Jayne, J.T., Ng, N.L. Investigating the Sources of Urban Air Pollution Using Low-Cost Air Quality Sensors at an Urban Atlanta Site. *Environmental Science & Technology*, 56(11), 7063–7073, 2022.
https://doi.org/10.1021/acs.est.1c07005

[2] **Yang, L.H.**, Takeuchi, M., Chen, Y., Ng, N.L. Characterization of Thermal Decomposition of Oxygenated Organic Compounds in FIGAERO-CIMS. *Aerosol Science and Technology*, 55(12), 1321–1342, 2021.
https://doi.org/10.1080/02786826.2021.1945529

[1] **Yang, L.H.**, Han, B.A. Data-driven Predictions and Novel Hypotheses about Zoonotic Tick Vectors from the Genus Ixodes. *BMC Ecology*, 18(7), 1–6, 2018.
https://doi.org/10.1186/s12898-018-0163-2

Co-Authored Papers

[20] Lucille G.G., **Yang L.H.**, Mickley L.J., Oak Y.J. Comparing rural and urban NO2 column densities from GEMS across East Asia: Diurnal and seasonal differences, *Atmospheric Environment*, 382, 122176, 2026.
https://doi.org/10.1016/j.atmosenv.2026.122176

[18] Dang R., Jacob D.J., Wang H., Nowlan C.R., Abad G.G., Chong H., Liu X., Shah V., **Yang L.H.**, Oak Y.J,, Marais E.A., Horner R.P., Rollins A.W., Crawford J.H., Li K., and Liao H. High-resolution Geostationary Satellite Observations of Free Tropospheric NO_2 over North America and Implications for Lightning Emissions, *Proceedings of the National Academy of Sciences*, 122(42), e2510535122, 2025.
https://doi.org/10.1073/pnas.2510535122

[17] Colombi N.K., Jacob D.J., Ye X., Yantosca B., Bates K., Pendergrass D.C., **Yang L.H.**, Li K., and Liao H. Large and Increasing Stratospheric Contribution to Tropospheric Ozone over East Asia, Under review by *Atmospheric Chemistry and Physics*, 2025.

[16] Pendergrass, D.C., Jacob, D.J., Oak, Y.J., Dang, R., **Yang, L.H.**, Beaudry, E., Colombi, N.K., Zhai, S., Kim, H., Choi, J.S., Park, J.S., Kim, S., Li, K., Liao, H. Wintertime Trends of Fine Particulate Matter

($PM_{2.5}$) in South Korea, 2021 - 2022: Response of Nitrate and Organic Components to Decreasing NO_x Emissions. *Geophysical Research Letters*, 52(19), e2025GL116091, 2025.
https://doi.org/10.1029/2025GL116091

[15] Westgate, S., Shivji, Z., **Yang, L.H.**, Rivera-Rios, J.C., Ng, N.L. Field evaluation of low-cost particulate matter (PM) sensor instruments in indoor and outdoor environments: A university lecture hall and urban southeastern United States *Aerosol Science and Technology*, 1–22, 2025.
https://doi.org/10.1080/02786826.2025.2484244

[14] Beaudry, E., Jacob, D.J., Bates, K., Zhai, S., **Yang, L.H.**, Pendergrass, D.C., Colombi, N., Simpson, I., Wisthaler, A., Hopkins, J., Ke, L., Liao, H. Ethanol and Methanol in South Korea and China: Evidence for Large Anthropogenic Emissions Missing from Current Inventories. *ACS ES&T Air*, 2(4), 456-465, 2025.
https://doi.org/10.1021/acsestair.4c00210

[12] Dang, R., Jacob, D.J., Zhai, S., **Yang, L.H.**, Pendergrass, D.C., Coheur, P., Clarisse, L., Van Damme, M., Choi, J., Park, J., Liu, Z., Xie, P., Liao, H. A Satellite-Based Indicator for Diagnosing Particulate Nitrate Sensitivity to Precursor Emissions: Application to East Asia. *Environmental Science & Technology*, 58(45), 20101–20113, 2024.
https://pubs.acs.org/doi/10.1021/acs.est.4c08082

[11] Lin, H., Emmons, L.K., Lundgren, E.W., **Yang, L.H.**, Feng, X., Dang, R., Zhai, S., Tang, Y., Kelp, M.M., Colombi, N.K., Eastham, S.D., Fritz, T.M., Jacob, D.J. Intercomparison of GEOS-Chem and CAM-chem Tropospheric Oxidant Chemistry within CESM2. *Atmospheric Chemistry and Physics*, **24**(15), 8607–8624, 2024.
https://doi.org/10.5194/egusphere-2024-470

[10] Oak, Y.J., Jacob, D.J., Balasus, N., **Yang, L.H.**, Chong, H., Park, J., Lee, H., Lee, G.-T., Ha, E.-J., Park, R.J., Kwon, H.-A., Kim, J. A Bias-Corrected GEMS Satellite Product for NO_2 Using Machine Learning with TROPOMI. *Atmospheric Measurement Techniques*, 17(17), 5147–5159, 2024.
https://doi.org/10.5194/amt-17-5147-2024

[8] Varon, D., Jervis, D., Pandey, S., Gallardo, S., Balasus, N., **Yang, L.H.**, Jacob, D. Quantifying NO_x Point Sources with Landsat and Sentinel-2 Observations of NO_2 Plumes. *Proceedings of the National Academy of Science*, 121(27), e2317077121, 2024.
https://doi.org/10.1073/pnas.2317077121

[7] Westervelt, D., Isavulambire, P., Yombo Phaka, R., **Yang, L.H.**, Raheja, G., Milly, G., Selenge, J.-L., Mulumba, J.P., Bousiotis, D., Djibi, B., McNeill, V.F., Ng, N.L., Pope, F., Mbela, G., Konde, J. A Low-Cost Investigation into Sources of $PM_{2.5}$ in Kinshasa, DRC. *ACS ES&T Air*, 1(1), 43–51, 2023.
https://doi.org/10.1021/acsestair.3c00024

[6] Zhai, S., Jacob, D.J., Pendergrass, D., Colombi, N., Shah, V., **Yang, L.H.**, Zhang, Q., Sun, Y., Wang, S., Luo, G., Yu, F., Woo, J.-H., Kim, Y., Kim, H., Dibb, J.E., Lee, T., Han, J.-S., Li, K., Liao, H. Coarse Particulate Matter Air Quality in East Asia: Implications for Fine Particulate Nitrate. *Atmospheric Chemistry and Physics*, 23(7), 4271–4281, 2023.
https://doi.org/10.5194/acp-23-4271-2023

[5] Colombi, N.K., Jacob, D.J., **Yang, L.H.**, Zhai, S., Shah, V., Grange, S.K., Yantosca, R.M., Kim, S., Liao, H. Why Is Ozone in South Korea and the Seoul Metropolitan Area So High and Increasing? *Atmospheric Chemistry and Physics*, 23(7), 4031–4044, 2023.
https://doi.org/10.5194/acp-23-4031-2023

REPORT

Yang, L.H., Kim J, Ahn D.H., 2021. Air pollution: an introduction to its causes, effects, and solutions. Ministry of Foreign Affairs of the Republic of Korea, National Council on Climate and Air Quality of the Republic of Korea. Government Publications Registration Number: 11-1262000-000285-01.

PRESENTATIONS

[20] Evolving Sensitivity of $PM_{2.5}$ to Emission Controls in a Changing Atmospheric Chemical Regime. The 12th International GEOS-Chem Meeting, St. Louis, MO, June 2026. (Talk)

[19] Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. Methane Emissions Technology Alliance (META) led by Stanford University and Lawrence Berkeley National Laboratory, Remote, February 2026. (Invited Talk)

[18] Megacities as Indicators of National NO_x and CO_2 Emission Trends. AGU Annual Meeting, New Orleans, LA, USA, December 2025. (Talk)

[17] Advancing Chemical Transport Modeling for Air Quality, Satellite Retrievals, and the Clean Energy Transition. Center for Aerosol and Engineering (CASE) seminar, St. Louis, MO, USA, November 2025. (Talk)

[16] Modeling of Methyl Hydroperoxide Observations in Urban and Remote Air over South Korea: Methylperoxy Radical Chemistry and Inference of Atmospheric Methanediol. Caltech Laboratory for the Study of Atmospheric Composition Group Meeting, Pasadena, CA, USA, June 2025. (Invited Talk)

[15] Co-evolution of Fossil Fuel CO_2 and NO_x in Megacities around the World. NASA JPL Earth Action for Air Quality Meeting, La Cañada Flintridge, CA, USA, May 2025. (Talk)

[14] Advancing Chemical Transport Modeling for Air Quality, Satellite Retrievals, and the Clean Energy Transition. NASA JPL Tropospheric Composition Group Meeting, La Cañada Flintridge, CA, USA, May 2025. (Talk)

[13] Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. EGU Annual Meeting, Vienna, Austria, April 2025. (Talk)

[12] Modeling of methyl hydroperoxide observations in urban and remote air over South Korea: Demonstration of high-NO and low-NO regimes for hydrocarbon oxidation, and inference of atmospheric methanediol. AMS Annual Meeting, New Orleans, LA, USA, January 2025. (Talk)

[11] Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. AGU Annual Meeting, Washington, DC, USA, December 2024. (Talk)

[10] Interpreting GEMS Geostationary Satellite Observations of the Diurnal Variation of Nitrogen Dioxide (NO_2) over East Asia. The 104th AMS Annual Meeting, Baltimore, MD, USA, January 2024. (Poster)

[9] Interpreting GEMS Geostationary Satellite Observations of the Diurnal Variation of Nitrogen Dioxide (NO_2) over East Asia. The 14th GEMS Science Team Meeting, Jeju, South Korea, September 2023. (Talk)

[8] Interpreting GEMS Geostationary Satellite Observations of the Diurnal Variation of Nitrogen Dioxide (NO_2) over East Asia. NASA and EPA Joint Atmospheric Chemistry Group Meeting, Remote, December 2023. (Invited Talk)

[7] Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from Space. Joint Science Meeting for TEMPO, GeoXO, & TOLNet, Huntsville, AL, USA, May 2023. (Poster)

[6] Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from Space. Atmospheric Measurement Group Meeting, Center for Astrophysics – Harvard & Smithsonian, Cambridge, MA, USA, December 2023. (Invited Talk)

[5] Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from Space. AGU Fall Meeting, Chicago, IL, USA, December 2022. (Poster)

[4] Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from Space. The 13th GEMS Science Team Meeting, Seoul, South Korea, November 2022. (Talk)

[3] Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from

Space. The 10th International GEOS-Chem Meeting, St. Louis, MO, June 2022. (Talk)

[2] Understanding the Sources of Urban Air Quality Using Low-Cost Air Quality Sensors. The 39th AAAR Conference, Remote, October 2021. (Talk)

[1] Thermal Decomposition Characterization of Filter Inlet for Gases and AEROsols (FIGAERO) coupled with Chemical Ionization Time-of-Flight Mass Spectrometer (ToF-CIMS). The 37th AAAR Conference, Portland, OR, October 2019. (Poster)

MENTORSHIP

Jaden Southern (Stanford University, class of 2026) Summer 2024
Co-mentor with Ruijun Dang, Yujin Oak, and Loretta Mickley.
Project: Evaluating TEMPO Satellite HCHO and NO_2 Products Against Ground-Based Pandora Instruments.

Lucy Gagnon (Williams College, Class of 2024; now Ph.D. student at Duke University) Summer 2023
Co-mentor with Yujin Oak and Loretta Mickley.
Project: Spatial and Temporal Differences in NO_2 Column Densities and Implications for Geostationary Satellite Product Applications Across Asia.

TEACHING EXPERIENCE

Harvard University, Teaching Fellow — Cambridge, MA
EPS 200: Atmospheric Chemistry and Physics — Spring 2023, Spring 2024

Georgia Institute of Technology, Teaching Assistant — Atlanta, GA
CEE 2300: Environmental Engineering Principles — Spring 2018, Spring 2019

Georgia Institute of Technology, Grader — Atlanta, GA
ME 3322: Thermodynamics — Spring 2019

SERVICE

Reviewer *Atmospheric Measurement Techniques; npj Clean Air; Environmental Science: Atmospheres*
NASA review panel (2025)

Leader *Group Life and Community Subgroup*, Harvard Atmospheric Chemistry Modeling Group (ACMG) (2024 - 2025)

Organizer *Journal Club*, Harvard ACMG (2022–2024)
Graduate Student Forum, Harvard ACMG (2022–2024)
Undergraduate Research Symposium 2024, Harvard ACMG (2024)

Member American Geophysical Union, American Meteorological Society, Sigma Xi

Volunteer *Science Club for Girls*, Boston (2021)